

Analytic Geometry

Module 6 Test

Grade 9 • MTH1W

Time: 60 minutes • Total: 70 marks

Name: _____

Date: _____

Instructions: Show all work for full marks. Marks for each question are shown in brackets. Calculators are permitted unless your teacher states otherwise.

Section A — Knowledge & Understanding

1. State the slope of the line $y = 7x + 2$. **[1]**

2. State the slope of the line $y = -x + 4$. **[1]**

3. State the y-intercept of the line $y = 5x - 8$. **[1]**

4. State the slope of a line parallel to $y = 6x + 1$. **[1]**

5. State the slope of a line perpendicular to a line with slope 2. **[1]**

6. Find the midpoint of (0, 0) and (4, 10). **[1]**

7. Find the length of the segment from (0, 0) to (0, 7). **[1]**

8. Find the slope of the line through (0, 0) and (3, 9). **[1]**

9. Does the point $(0, 3)$ lie on the line $y = 2x + 3$? **[1]**

10. Find the slope of the line through $(1, 2)$ and $(5, 14)$. **[2]**

11. Find the length of the segment from $(2, 1)$ to $(5, 5)$. **[2]**

12. Find the midpoint of $(-2, 4)$ and $(6, 10)$. **[2]**

13. State the slope of a line perpendicular to $y = -3x + 2$. **[2]**

14. Write the equation of the line parallel to $y = 2x - 1$ through $(0, 5)$. **[2]**

15. Find the x-intercept of the line $y = -x + 4$. **[2]**

Section B — Application

16. Find the slope of the line through $(-2, 5)$ and $(4, -7)$. **[2]**
17. Find the length of the segment from $(-3, 1)$ to $(2, 13)$. **[2]**

18. Find the midpoint of $(-5, -2)$ and $(7, 8)$. **[2]**
19. Determine whether $y = 4x + 1$ and $y = 4x - 9$ are parallel. **[2]**
20. State the slope of a line perpendicular to $y = (3/4)x + 2$. **[2]**
21. Find the x-intercept and y-intercept of $y = 3x - 12$. **[2]**
22. The line through $(2, 3)$ and $(8, k)$ has slope 2. Find the value of k . **[3]**
23. Rewrite $2x + y = 6$ in the form $y = mx + b$, then state the slope and y-intercept. **[3]**

24. A triangle has vertices $A(0, 0)$, $B(9, 0)$, $C(9, 12)$. Find its perimeter. **[3]**

Section C — Thinking

25. A line passes through $(1, 2)$ and $(4, 8)$. Find its slope, then the slope of any line perpendicular to it. **[4]**
26. A line passes through $(0, -3)$ and $(2, 5)$. Write its equation, then state whether it is parallel to $y = 4x + 7$. **[4]**

27. The midpoint of segment AB is $M(4, 3)$. If $A = (1, -1)$, find the coordinates of B, then find the length of AB. (Very hard.) **[5]**
28. A line is perpendicular to $y = (1/2)x + 4$ and passes through $(0, 5)$. Write its equation, then find its x-intercept. (Very hard.) **[5]**
29. A triangle has vertices $P(-2, -1)$, $Q(4, -1)$, $R(4, 7)$. Find the length of each side and the perimeter, and show the triangle is right-angled. (Very hard.) **[6]**

Section D — Communication

30. A student claims that $y = 2x + 1$ and $y = 2x + 6$ must eventually cross because "all straight lines meet somewhere." Using slope, explain why the student is incorrect. (Very hard.) **[4]**

— END OF EXAM —